	the collision is inelastic.
3(a)	A gravitational field is a region of space in which a test mass experiences an attractive gravitational force in the direction of the gravitational field, due to the attraction of other masses. The strength of the field is defined as the force per unit mass on a small test mass.
3(b)(i)	Gravitational force provides the centripetal force required

	$\dfrac{GMm_{\text{sun}}}{r^2} = \dfrac{m_{\text{sun}}v^2}{r}$ $\dfrac{GM}{r} = v^2$ $v = \sqrt{\dfrac{GM}{r}}$ (shown)
3(b)(ii)	Using $\dfrac{GM}{r} = v^2$, $M = \dfrac{rv^2}{G}$ $= \dfrac{\left(2.4\times10^{20}\right)\left(230\times10^{3}\right)^2}{6.67\times10^{-11}}$ $= 1.903\times10^{41}$ kg Ratio $= \dfrac{1.903\times10^{41}}{2.0\times10^{30}}$ $= 9.52\times10^{10}$ $\approx 9.5\times10^{10}$
4(a)	